



UNIVERSITY OF SALERNO

Department of Computer Science
Bachelor's Degree in Computer Science

BACHELOR THESIS

A Modern Edge-AI Prototype for Secure Smart Environments

Design, implementation, and experimental validation

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ISISLab

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Laboratory ISISLab

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Keywords edge AI; cybersecurity; IoT; experimental evaluation

Abstract

This bachelor thesis presents a compact edge-AI prototype for secure smart environments. The demo illustrates the complete path from a measurable requirement to a modular implementation and an experimental evaluation. The template uses an almost-white page, ISISLab blue accents, modern information cards, and subtle transparent laboratory marks. Replace this text with a concise account of the problem, method, implementation, results, limitations, and conclusions.

Keywords: edge AI; cybersecurity; IoT; experimental evaluation

Declaration

I declare that this thesis is my original work and that all sources have been acknowledged. Adapt this statement to the official University of Salerno requirements and to the regulations of the degree programme.

Place and date: _____ Signature: _____

Acknowledgements

This optional page demonstrates the visual treatment of acknowledgements. Replace it with a brief and personal text, or remove the command from the project when no acknowledgement page is required.

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1. Introduction

1.1. Motivation

Intelligent connected devices are increasingly deployed in homes, laboratories, and industrial spaces. Their usefulness grows with their ability to interpret data locally, yet resource constraints and security requirements make design decisions non-trivial [1, 2].

Research question

?

How can a compact edge device deliver useful inference performance while preserving responsiveness and a clear security boundary?

1.2. Objectives

The bachelor project follows four practical objectives:

- define a measurable system requirement;
- implement a modular prototype;
- evaluate latency, accuracy, and resource use;
- discuss limitations and future improvements.

Template guidance

Use the opening chapter to establish context, define the problem, state objectives, and preview the thesis structure. Avoid implementation details that belong in later chapters.

1.3. Document structure

Chapter 2 presents the design, Chapter 3 reports the evaluation, and the final chapter summarises the main findings.

2. System Design

2.1. Architecture

The demonstrator separates sensing, inference, policy, and presentation. Figure 2.1 uses only native LaTeX boxes, so it compiles quickly and can be replaced with a project-specific image.



Figure 2.1.: Lightweight demo architecture for the bachelor thesis.

2.2. Implementation notes

Describe hardware, software modules, interfaces, and key engineering choices. Keep reproducibility in mind by recording versions and configuration parameters.

```
def classify(sample, model, threshold=0.75):  
    score = model.predict(sample)  
    return int(score >= threshold), float(score)
```

Design decision

The prototype uses a modular pipeline so that the inference model and the security policy can be tested independently.

3. Experimental Evaluation

3.1. Protocol

Report the dataset or workload, train/test split, baseline, metrics, hardware, and repeated trials. The demo values below are placeholders.



Figure 3.1.: Illustrative evaluation result drawn with simple rules.

Table 3.1.: Example performance summary.

Configuration	Accuracy	Latency	Memory
Baseline	72.0%	84 ms	410 MB
Prototype	86.0%	41 ms	286 MB
Optimised	91.0%	29 ms	241 MB

Interpretation

Discuss effect size, uncertainty, practical relevance, threats to validity, and any discrepancy between expected and observed behaviour.



4. Conclusions

The project demonstrates a complete path from problem definition to implementation and evaluation. Summarise the contribution without repeating the abstract, then state limitations and concrete next steps.

4.1. Future work

Potential extensions include a broader dataset, additional threat models, deployment on alternative hardware, and a user study of the alert interface.

References

- [1] Rodrigo Roman, Javier Lopez, and Masahiro Mambo. 2018. Mobile Edge Computing, Fog et al.: A Survey and Analysis of Security Threats and Challenges. *Future Generation Computer Systems* 78 (2018), 680–698.
- [2] Weisong Shi, Jie Cao, Quan Zhang, Youhuizi Li, and Lanyu Xu. 2016. Edge Computing: Vision and Challenges. *IEEE Internet of Things Journal* 3, 5 (2016), 637–646.

A. Reproducibility Checklist

Before submission

Replace all sample metadata, verify the official University of Salerno cover wording, check every citation, and compile the bibliography with BibTeX.